Verifiable AI via the $\ell^1 CoboundaryNorm$

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Abstract

This paper specifies a model-agnostic verification framework for AI outputs grounded in the **structural consistency** of declared domain graphs. We establish that "hallucination elimination" and "truth verification" are ill-posed as universal mathematical problems; instead, we operationalize **domain-relative structural consistency certification**. By mapping AI outputs to sections of a finite constraint graph and solving the resulting ℓ^1 coboundary quotient problem, we generate auditable, replayable certificates of consistency. We prove that this certification is robust against "shortcut" evaluation through **Observer Passivity**, ensuring that downstream evaluators cannot promote unverified residuals.

The Operational Claim

The Structural Consistency Claim

domain-relative structural consistency certification

1 Scope: Constraint Audit Infrastructure

Earlier papers establish the algebraic foundations (Paper 009) and the governance of evaluators (Paper 011). This paper provides the **engineering bridge**, transforming abstract obstruction theory into a deployable audit infrastructure.

1.1 What is Being Certified

This verifier does not judge the "intelligence" of the model or the "truth" of the world. It provides a formal, reproducible certificate that an AI output respects the explicit constraints of a specified domain graph.

- **Measured:** Inconsistency relative to fibers, restriction maps, and local gauges.
- **Not Measured:** Hallucination elimination, universal truth, or cognitive states.

2 The Verification Boundary: What the Audit Measures

The verifier measures structural inconsistency relative to a specified domain graph. It is a post-hoc audit layer, not a predictive or generative model.

3 Primal-Dual Quotient Decomposition

Given a defect field $\delta \in C^1(G)$, define:

$$\Phi([\delta]) = \inf_{f \in C^0(G)} \|\delta - d_0 f\|_1$$

Here $d_0 f$ is the exact defect generated by an allowed gauge shift. The quantity $\Phi([\delta])$ is the cost of the best representative of the defect class modulo allowed repairs.

4 Exact LP Solver and Heuristic Repair

The exact quotient quantity is computed by the LP:

$$\min_{f,z} \sum_{e \in E} z_e \quad \text{subject to} \quad -z_e \leq \delta_e - (d_0 f)_e \leq z_e, \quad z_e \geq 0$$

5 Final Law

The Audit Consistency Law

Domain-graph consistency is a finite, reproducible, and auditable property of an AI output section relative to a declared constraint graph, certified via the ℓ^1 coboundary quotient LP.

A Implementation Registry

The following modules in the `omega_engine` implement the domain-graph verifier:

- **Coboundary Auditor:** `coboundary_auditor.rs`
- **LP Quotient Solver:** `quotient_solver.rs`
- **Audit Manifest Generator:** `audit_manifest.rs`

References

1. Carroll, J. H. (2026). *Primitive Axiomatics of Finite Cohomological Obstruction Theory* 2026. Zenodo. https://zenodo.org/records/20172395.
2. Carroll, J. H. (2026). *Degree-Wise Obstruction Calculus,* Zenodo.
3. Carroll, J. H. (2026). *Guarded Descent and Layered Authority,* Zenodo.